

MS-50 Modular

Design pack for a white-box modular FX VST

Personal study instrument. Not a Korg product.

Document date: 2026-09-21

Stack specified here: C++20, JUCE 8, CMake, VST3 effect, stereo in and stereo out, mono module graph, private git repository.

Revisions

2026-09-21: first compiled pack. Cover, methodology, research summary, software schematic, build guide, test plan, and bibliography.

A generated timestamp is not a revision. Later edits add a line here and a date on the changed section heading in the markdown.

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Methodology

Date of this packet: 2026-09-21.

The plugin is a white-box modular instrument. Each module has a stated signal path, a stated set of jacks, and a list of numbers that are either taken from the 1978 drawings or marked as stand-ins. A later revision replaces a stand-in by editing one constant and its note. It does not retrain a network.

Why white-box

The hardware was not available for measurement. What exists is a schematic set, one peer-reviewed analysis of the filter (Rest, Parker, Werner, DAFx 2017), and a handful of public recordings. That evidence describes topology, a few labeled voltages, and one filter circuit. It does not describe a corpus of paired input and output audio.

A neural model would fit those recordings, including the room, the other synths in the video, and the effects some of them print. It would also be free to invent a Korg-35 filter, one pitch law, and two ADSRs, which are the usual mistakes listed in the research. A modular FX plugin with animated cables needs separate modules anyway. The structure has to exist as objects, not as a single black box.

White-box here means: the phase 1 VCO is a digital oscillator with the confirmed control topology, not a SPICE netlist of the 2SC1583. The phase 1 filter starts as a 2-pole lowpass (step 8) and moves toward the diode bridge (step 20). Both stages are written down. Neither stage claims to be a transistor-level copy.

Source hierarchy

When two sources disagree, the higher one wins.

Rank	Source	Use
1	Korg drawings, 78-11-8 and 78-11-9, KOD-A40038 through A40049	Jack names, knob names, topology, parts that are printed (CA3019, RC4200, 4013, 555, 7815/7915, +18 V on the VCO, MG voltage cartoons, EG2 voltage cartoons, footage switch)
2	Rest, Parker, Werner, DAFx-17 paper 88	Filter small-signal structure, which parts of their model were simplified, the warning that 1N4148 parameters are substitutes
3	Public recordings and Alex Ball's panel survey	Listening checks and the VCA 1 / VCA 2 panel mapping
4	Inference in docs/01-research.md	Trigger polarity, MG modulation law, EG2 time order. Usable only where the research marks INFERRED
5	Stand-ins in this file	Plugin numbers that the drawings do not print

MS-20 owner's manual voltages are rank 4 at best and only as a warning. They are not copied onto the MS-50.

Panel mapping that is inferred, not printed: the sheet says VCA for the CA3019 audio VCA with LO CUT and INTENSITY, and MVCA for the optocoupler. This packet calls them VCA 1 and VCA 2 because that is the only published panel description that matches those knobs (Alex Ball). Code comments must say "schematic name VCA" and "schematic name MVCA".

Accuracy target

The target is a patched modular FX that a reader of the drawings would recognize:

- Confirmed topology is present: two VCO pitch jacks, scale switch on Hz/V only, simultaneous VCO waves, diode-bridge filter goal, two different VCAs, MG ranges and voltage cartoons, EG1 ADSR with three outs, EG2 without decay or sustain, divider by 2 and by 4 only, inverter gain -1, integrator as lag, ring mod as a four-quadrant multiply.
- Confirmed labels that have numbers are reproduced on the internal volt scale below.
- Unknowns are stand-ins, each with an id, a single constant, and a replacement test.

- Listening against public video is qualitative. There will be no null against a Gladén recording. Those files include an MS-20, a room, and unknown knob positions.

Out of target for steps 0 to 20: transistor VCO, measured Hz/V tracking error, CA3019 Shockley parameters from a datasheet (the paper says the datasheet does not give them), optocoupler lag in milliseconds, EG time in seconds per knob degree, noise spectrum in dB, buffered versus passive multiples.

Volt scale

Internal audio-thread values are volts, as `float`. The drawings that print a voltage use 5 V and 2.5 V. The plugin converts at the output stage only.

Hardware	Float volts in the graph	Float at the plugin output, after Output level
0 V	0	0
+5 V	5	+1.0 at default output level (S-01)
-5 V	-5	-1.0
MG triangle, confirmed ± 2.5 V	± 2.5	± 0.5
MG pulse, confirmed 0 to +5 V	0 to +5	0 to +1

S-01: `kOutputVoltsToUnit = 0.2` so that 5 V becomes 1.0 at the DAC when Output level is 1. This scale is a stand-in for jacks whose peak voltage was not printed. It is fixed by the MG and EG2 cartoons for those jacks. Replacement: if a scope shot shows the MG triangle is not ± 2.5 V at the jack, change the MG output gains, not this constant, unless the whole cartoon set was misread.

How stand-ins are marked

In documents: the id S-xx and the word STAND-IN.

In code, on the constant itself:

```
// STAND-IN S-03: 1 V on CV IN Hz/V produces the footage frequency.
// Replace after measuring Hz versus DC at each SCALE position.
inline constexpr float kHzPerVoltRef = 1.0f;
```

Rules:

- One id, one constant. Modules call the constant. They do not restate the number.
- A CONFIRMED quantity does not get a stand-in id. MG frequency endpoints 0.01 Hz and 200 Hz are confirmed. The pot taper between them is S-16.
- PAPER-SUBSTITUTE is a stand-in that came from the DAFx paper's 1N4148 fit (S-10). It is not a Korg number.
- Do not "improve" a stand-in by ear during a step whose acceptance test is functional. Ear changes get a new id and a note of what was listened to.

Stand-in register

ID	Quantity	Value used in the plugin	Replace when
S-01	DAC scale	5 V graph = 1.0 output unit at level 1	A confirmed jack voltage disagrees with the cartoons
S-02	VCO wave peaks	Saw, triangle, and pulse are bipolar ± 5 V	A scope shot of the three VCO jacks
S-03	Hz/V law	$f = \text{footageHz} * \max(\text{volts}, 0.05) / 1.0$	Measured Hz versus DC on CV IN Hz/V at each scale
S-04	OCT/V law and fine tune	0 V = footage Hz. +1 V = +1 octave. Fine tune ± 2 semitones	Measured OCT/V slope and fine-tune span

ID	Quantity	Value used in the plugin	Replace when
S-05	VCO FREQ A/B depth	Full attenuator, ± 5 V in, adds ± 5 octaves through the expo summer	Measured FM depth
S-06	VCO pulse width	Duty 0.05 to 0.95. Knob and PWM jack sum	Measured duty versus knob
S-07	VCF cutoff knob	Exponential 20 Hz to 18 kHz at 0 V CV, small input	Measured cutoff versus knob at low peak
S-08	VCF cutoff CV	Attenuator 0 to 1 scales a ± 5 V jack. Positive volts raise cutoff. Not V/octave and not Hz/V	Measured cutoff versus DC
S-09	VCF peak, step 8	Resonant 2-pole, Q from 0.5 to 8, stable at full peak	Step 20 replaces this row's Q law
S-10	Diode bridge, step 20	$r_d = n \cdot V_T / I_b$ with paper 1N4148 $I_s = 2.52e-9$, $n = 1.752$. Feedback clip is $\tanh$. Input level may pull cutoff down. Do not hard-code 250 Hz	A measured CA3019 fit, or a decision to drop the paper's I_s and n
S-11	VCA 1 gain	$gain = \text{clamp}(\text{envVolts}/5, 0, 1) * \text{intensity}$. Low-cut is a one-pole highpass, 10 Hz to 2 kHz	A diode-bridge gain law measured or derived from KOD-A40045 without new guesses
S-12	VCA 2 / MVCA	$gain = \text{clamp}(\text{cvVolts}/5, 0, 1)$ then a one-pole smoother, 20 ms	Measured optocoupler step response
S-13	EG time knobs	Exponential 1 ms to 10 s	A scope shot or a manual sentence
S-14	EG1 levels	OutA 0 to +5 V. OutB 0 to -5 V. OutC bipolar: peak +5 V, sustain 0.5 maps to 0 V	OutC is the weak one: the EG1 sheet did not print volts. OutA/OutB volts are also unprinted and follow EG2's 5 V cartoons only as a stand-in
S-15	Trigger threshold	A CV/audio trig jack is "held" when volts < 1.5. Unpatched Trig rests at +5 V (released). A Gate held promotes to 0 V. A Gate released promotes to +5 V	Measured EG trig threshold
S-16	MG pot curves	Frequency knob exponential from 0.01 Hz to 200 Hz. One PW knob sets triangle symmetry and pulse width together	The endpoints are confirmed. The curve and the symmetry map are not
S-17	Noise level	White and pink bipolar, peak about ± 2.5 V, fixed internal trim	Measured jack level and spectrum
S-18	Divider thresholds	Schmitt: high at +0.5 V, low at +0.3 V. Outputs are 0 V or +5 V, square	Measured comparator threshold and pulse height
S-19	Inverter offset	Offset forced to 0. The sheet's trim exists and is not a panel control	A unit with a large residual offset
S-20	Integrator time	One-pole lag, tau 1 ms to 2 s, same sign as the input. A held input settles at that voltage	Measured glide time, and a check that a held DC input does not ramp away
S-21	Ring scale	$\text{outVolts} = (\text{aVolts} * \text{bVolts}) / 5$. Bleed constant 0	Measured full-scale product and residual feedthrough
S-22	Ext In gate	Envelope of $\text{abs}(\text{mono})$ with an attack/release follower. Fires above a threshold knob. Manual button ORs in a held gate	Phase 2 ESP trigger and TRIG SW replace this. The button is not a separate module
S-23	Output mix law	$y = \text{dry} * (1 - \text{mix}) + \text{wet} * \text{mix}$, wet mono copied to L and R, then $* \text{level} * 0.2$	Only if the headphone amp is ever modeled, which is phase 2
S-24	Feedback edge	In a cycle, the newest cable is the back-edge and delays one sample	A different deterministic rule, if documented in the same change as the tests

ID	Quantity	Value used in the plugin	Replace when
S-25	Control rate	Knobs and CV are read every audio sample. There is no slower CV block	A profiled need for a coarser rate. Do not add one for style
S-26	EG2 timeline	See SCHEMATICS, EG2. No sustain plateau	An owner's sentence or a scope shot of hold versus delay
S-27	Output fan-out	One output may feed many inputs. One input accepts one cable	If a later pass removes fan-out because the hardware jacks do not stack, the multiples module (phase 2) becomes required first

Footage frequencies used with S-03 and S-04, equal-tempered C, A440. These pitches are a stand-in for what "32'" meant at the jack with no cable inserted. The switch positions 32', 16', 8', 4' are confirmed. The hertz numbers are not.

SCALE	Stand-in Hz at the reference
32'	32.703
16'	65.406
8'	130.813
4'	261.626

How to A/B against public recordings

Do this after step 8 (filter stand-in) and again after step 20 (diode-bridge pass). Do not treat adjectives as data. One listener called the MS-50 harsher than his MS-20. Another called it softer.

Procedure:

1. Play the reference video from the start time in docs/TESTPLAN.md. Note only what the uploader claimed about the patch (saw, peak down, peak up, filter alone, Hz/V, S-trig, effects on or off).
2. Set the plugin to the same claim. Unknown knobs stay at the packet defaults. Write the knob values into the test note.
3. Listen for the structural checks in the test plan: peak changes the tone, full peak can ring, a hot input in step 20 moves more than the level, Hz/V and OCT/V do not track the same keyboard, EG2 has no sustain flat-top, MG pulse is unipolar.
4. Record the plugin dry from the host, no extra plug-ins. Store that recording outside the repo (see docs/REPO_SETUP.md).
5. Write three lines: what matched, what did not, which stand-in id would have to move. Do not retune S-numbers in the same session as the listening. A second change, with the id edited and the test note linked, is the retune.

Files that are not dry enough for timbre: Dr. Kunz (Small Stone, SDD-3000, dbx) and the Perfect Circuit jam (MS-20 in the same mix). They are patch ideas only.

Definition of done for a module

A module is done when all of the following are true:

1. Its jacks, knobs, and confirmed topology match docs/SCHEMATICS.md.
2. Every numeric choice is either confirmed or an id in the register above.
3. The automated test named in the build-guide step passes.
4. Unpatched inputs read 0 V, except Gate promotion tests.
5. `processSample()` does not allocate. A test or a code search shows no `new`, `vector::push_back`, `string`, `lock`, or `printf` on that path.
6. The module is on the rack as its own faceplate. The integrator is not hidden inside another module.
7. A wrong cable (second cable into an input, or a signal into a Gate-only jack) is rejected by the graph, not only hidden by the UI.

A module is not done because it "sounds analog."

Definition of done for the phase 1 packet

Steps 0 through 20 in docs/BUILD_GUIDE.md have each been accepted. Default patch loads. Dry mix 0 passes stereo. Step 20 filter is the one a listener A/Bs, and the note from that listen exists, including anything that still disagrees.

Phase 2 remains unbuilt.

Research summary

Date: 2026-09-21.

This file is the evidence the software schematic is allowed to use. It condenses the study of the Korg MS-50 drawings and the DAFx 2017 filter paper. It is not a second schematic of the PCB.

Status words:

- CONFIRMED: printed or drawn on the November 1978 sheets, or stated by the DAFx paper as a property of those sheets.
- INFERRED: a reading of a drawing, or a later panel description.
- UNKNOWN: not printed, and not settled by the paper.

Plugin numbers for UNKNOWN items live in METHODOLOGY.md as stand-ins. They are not repeated here as if they were measured.

What was opened

Document	What it actually is
z3sound Korg MS-50 Schematics.pdf	12 image pages. Drawing dates 78-11-8 and 78-11-9. Boards KLM-164 / KLM-164B and KLM-165B. Title blocks KOD-A40038 through A40049
Archive.org Korg_MS-50_Service_Manual.pdf inside korg_service_manuals	Another 12-page image set with the same page sizes. Not a written calibration procedure
Rest, Parker, Werner, DAFx-17 paper 88	Analysis and a simplified WDF of the VCF. Their reference [15] is "KORG, Model MS-50 Circuit Diagram, Nov. 1978"

No MS-50 owner's manual was found. MS-20 owner's manual (Archive.org JL11275) is family context for S-trig and for what not to copy.

No cable on these sheets runs from one module's output to another module's input. A voice exists only if it is patched. CONFIRMED.

Power, KOD-A40038: regulators 7815 and 7915, so +15 V and -15 V, plus a +18 V node ahead of the 7815. The VCO integrator is drawn from +18 V. Transformer printed as KA-331 or KB-331. Fuse 0.5 A. Which transformer is which mains voltage is UNKNOWN.

Pitch and trigger

Path	Evidence	Status
CV IN Hz/V	Linear chain of eight 100 k Ω 0.1% resistors. SCALE 4', 8', 16', 32' sits on this chain only	CONFIRMED
CV IN OCT/V	Separate jack into an exponential pair. R1 marked 1 k Ω , +3000 ppm/ $^{\circ}$ C	CONFIRMED
FREQ A, FREQ B	Attenuated inputs drawn into the same summer as OCT/V	CONFIRMED topology. Volts per octave of these jacks: UNKNOWN
VCF CUTOFF FREQ	Attenuator into the diode bias summer. No Hz/V or OCT/V label	CONFIRMED unlabeled
EG TRIG IN	Diode and transistor network. No threshold printed	Voltage UNKNOWN. S-trig (active short to ground) is INFERRED
TRIG SW	Rests pulled up. Button pulls the output toward ground through 2.2 k Ω	CONFIRMED active-low. Phase 2 module

MG pulse cartoon: 0 to +5 V. MG triangle cartoon: ± 2.5 V. EG2 cartoons: 0 to +5 V on one jack and 0 to -5 V on the other. CONFIRMED annotations. VCO, VCF, VCA, noise, divider, and EG1 peak voltages are UNKNOWN. Do not

paste MS-20 "5 V peak-to-peak" or "2OCT/V" onto them.

Phase 1 modules

VCO (KOD-A40044)

Knobs: SCALE (4', 8', 16', 32'), FREQ A, FREQ B, FINE TUNING, PW/PWM. ADJ is an internal trim.

Jacks: CV IN Hz/V, CV IN OCT/V, two FM inputs, PWM, and three waveform outputs drawn with no series output capacitor. Cartoons: pulse, saw, triangle, simultaneous. No sync jack.

Quirks that the plugin must keep: footage switch does not sit on the OCT/V jack. FM jacks are not a second Hz/V input. Core supply is +18 V, which the digital model does not simulate. Wave amplitudes UNKNOWN (stand-in S-02).

VCF (KOD-A40045)

CA3019 (six matched diodes) biased as controlled resistances inside a Sallen-Key lowpass. PEAK is VR10, 10 k Ω , in the feedback. Eight discrete diodes D5 to D12 clip that feedback. SIG IN meets R59 = 100 k Ω , a unity buffer, then capacitive coupling into the bridge (paper, describing the sheet). SIG OUT is AC-coupled. Manual cutoff, external cutoff, and a temperature term become opposite bias voltages.

This is the circuit used in the Korg 700, 700S, 800DV, and 900PS, patent US 4,039,980 (paper). It is not the MS-20 Korg-35 and not the later MS-20 OTA.

The paper's WDF omitted D5 to D12 on purpose, replaced the bias op-amps with 50 Ω sources, and used 1N4148 parameters ($I_s = 2.52e-9$, $n = 1.752$) because the CA3019 datasheet did not supply them. Those diode numbers are PAPER-SUBSTITUTE (S-10), not Korg values. In that simplified model, a normalized resonance of 0.85 self-oscillated near 250 Hz and fell toward 200 Hz as a slow input ramp reached 2 V, and the oscillation nearly died. Those hertz figures are one experiment. Do not hard-code them.

Parts that appear on the sheet and in the paper's Table 1: R44 220 k Ω , R45 1.8 k Ω , R46 100 Ω , R48 2.7 k Ω , R50 2.2 k Ω , R52 47 k Ω , R54 47 k Ω , R56 1 k Ω , C11 and C12 22 nF, C15 2.7 nF, C17 10 μ F, C52 220 pF, PEAK pot 10 k Ω . $R_s = 50 \Omega$ is the paper's substitute.

Cutoff knob hertz span: UNKNOWN (S-07). Cutoff CV law: diode bias, not a keyboard standard (S-08).

VCA 1, sheet name VCA (KOD-A40045)

Second CA3019 as the gain cell. ENV IN feeds the bias summer. LO CUT (VR14) is a manual highpass in front of the cell, with no CV jack. Alex Ball's video states the same thing: non-resonant, manual only. INTENSITY (VR15, 100 k Ω A) is drawn after the bridge as an output attenuator, not as a CV-depth knob. OUT is AC-coupled. No initial-gain jack. Closed-state feedthrough is an internal trim (VR13).

Panel number "VCA 1" is INFERRED. Gain law in volts is UNKNOWN (S-11).

VCA 2, sheet name MVCA (KOD-A40043)

Optocoupler marked HTV between two 4558 stages. CONTROL IN drives the lamp transistor. No panel knob. The signal path as drawn has no series coupling capacitor, so it can pass CV. Lamp lag is real and UNKNOWN (S-12). Panel number "VCA 2" is INFERRED.

MG (KOD-A40048)

The sheet says MG, not LFO. 555 core. Printed frequency text: 0.01 Hz to 200 Hz. Outputs: triangle ± 2.5 V, two saws of opposite slope annotated ± 2.5 V, pulse 0 to +5 V. One PW/PWM pot is wired into both the triangle symmetry network and the pulse comparator. FREQ MOD enters the timing current. A linear-Hz reading of that jack is INFERRED. The pot curve between 0.01 Hz and 200 Hz is a stand-in (S-16). The top of the range is audio, so MG is a second oscillator as well as an LFO.

EG 1 ADSR (KOD-A40046)

Knobs: attack, decay, sustain, release. One TRIG IN. Three output jacks with different cartoons. Voltages are not printed (S-14). This is not the MS-20 EG1, which is delay-attack-release.

EG 2 HDAR (KOD-A40047)

Knobs printed: HOLD, DELAY, ATTACK, RELEASE. No decay knob. No sustain knob. Jacks: TRIG IN, two envelope outs annotated 0 to +5 V and 0 to -5 V, and DELAY TRIG OUT. Left-to-right drawing order is hold, then delay, then the attack/release slope generator. The musical definition of hold versus delay is INFERRED (S-26). Cartoons are bumps, not a sustain shelf.

Noise (KOD-A40042)

Selected transistor marked CG44R. W.NOISE before a passive filter. P.NOISE after it (0.022 μ F and 220 k Ω are readable in that branch). No panel knob. Internal LEVEL ADJ. Both outputs AC-coupled. Peak voltage UNKNOWN (S-17).

Divider (KOD-A40041)

IN to an LM339, then a 4013 as two toggle stages. Outputs labeled $\div 2$ and $\div 4$. No $\div 8$. Output amplitude UNKNOWN (S-18). Threshold UNKNOWN.

Inverter (KOD-A40043)

4558, input and feedback both drawn as 100 k Ω metal film, so the drawn gain is -1. DC-coupled. Offset trim VR2 is internal, not a panel knob (S-19).

Integrator (KOD-A40039)

IN, TIME, OUT. Op-amp lag: a held input must settle at the input voltage, same sign. It is not a resonant filter and not a ramp that runs away. Tau versus knob is UNKNOWN (S-20). The module stays on the rack.

Ring (KOD-A40040)

RC4200 four-quadrant multiplier. A IN, B IN, OUT. DC-coupled. Null trims VR7, VR8, VR9 are internal. No panel knob. This is not the filter's CA3019. Full-scale product is UNKNOWN (S-21). Using one input as a unipolar CV makes it a VCA. That patch is documented by a player (Dr. Kunz) and is allowed by a four-quadrant multiply.

Phase 2, on the same sheets, not in steps 0 to 20

Block	Sheet	Keep for later
Adding amplifier	KOD-A40049	Three level pots, inverting summer, DC-coupled. 220 k Ω into 330 k Ω feedback as drawn. No published gain spec
Sample and hold	KOD-A40040	IN, OUT, EXT CLOCK IN, CLOCK OUT, CLOCK pot, 555, K30(GR) FET, 3140 buffer, 0.1 μ F hold cap drawn
Audio amplifier (ESP)	KOD-A40042	SG IN with level, SG OUT, envelope follower, peak LED, trig LED, TRIG OUT. No bandpass. No frequency-to-voltage converter
TRIG SW	KOD-A40041	Active-low momentary. Phase 1 uses Ext In's button as stand-in S-22, not this module
Volt supply	KOD-A40039	Pot between the rails, 3.3 k Ω in each leg. Soft bipolar CV. Do not label it ± 5 V or ± 10 V
Meter	KOD-A40039	AC.IN, DC.IN, movement YL-80S
Headphone amp	KOD-A40038	SIG IN, level, PHONES. No separate line-out jack is drawn on this sheet
Multiples	absent	Alex Ball lists buffered multiples. Buffering is UNKNOWN. No circuit is on the 12 pages

Recordings worth using

Item	Use
Magnus Gladén, "Korg MS-20 vs. Korg MS-50 low pass filter test" (YouTube MwJk36KnB04)	Best filter compare. Saw, peak down, peak up, filter alone
Synth Party, MS-50 sequenced by a Yamaha CS-30 (YouTube C6oqCY0ArvM)	After 0:51 the uploader says the line is direct, driven by Hz/V and S-trig
Alex Ball, "The Rare KORG MS-50 from 1978" (YouTube 3YFDDDXw6F8), filter talk near 7:41 and compare near 8:43	Diode ring versus type 35, and the spoken low-cut claim. Musical, not a lab sweep
vitalance, SQ-10 / MS-50 / KR-55 (YouTube TZKxrRzoDU0)	Trigger and cutoff motion. Other gear is present
Dr. Kunz, MS-50 Textures (YouTube _iJJYswUYd0)	Ring used as a VCA. Effects are on the file. Not a timbre reference
Perfect Circuit MS-20 and MS-50 jam (YouTube UXW39L0-bxY)	Not isolated

Bibliography

- Korg MS-50 schematics, drawings 78-11-8 and 78-11-9.
<https://z3sound.com/manuals/Korg/Korg%20MS-50%20Schematics.pdf>
- Archive.org image set: <https://archive.org/details/synthmanual-korg-ms-50-schematics>
- Archive.org file titled service manual, same kind of image set:
https://archive.org/download/korg_service_manuals/KORG_ALL_SM/Korg_MS-50_Service_Manual.pdf
- Synthfool index of other MS-series files: https://synthfool.com/docs/Korg/MS_series
- Maximilian Rest, Julian D. Parker, Kurt James Werner, "WDF Modeling of a Korg MS-50 Based Non-Linear Diode Bridge VCF," DAFx-17, Edinburgh, 5 to 9 September 2017.
https://www.dafx.de/paper-archive/2017/papers/DAFx17_paper_88.pdf
- Yasuji Nagahama, US 4,039,980, "Voltage-controlled filter," issued 2 August 1977.
<https://patents.google.com/patent/US4039980A/en>
- Tim Stinchcombe, "A study of the Korg MS10 & MS20 filters" (2006). Contrast only.
http://www.timstinchcombe.co.uk/synth/MS20_study.pdf
- Alex Ball, KORG MS Series panel notes. <https://www.alexballmusic.com/korg-ms-series>
- Korg MS-20 owner's manual, Archive.org JL11275. <https://archive.org/details/JL11275>
- Sequencer.de MS50 page. Useful as an example of the "two ADSRs" error.
<https://www.sequencer.de/syns/korg/MS50.html>
- AMSynths AM8319. A later Eurorack reading that adds a highpass the MS-50 VCF sheet does not show.
<https://modulargrid.net/e/amsynths-am8319>

Software schematics

Date: 2026-09-21.

This is the signal path of the plugin. It is not a redraw of the Korg PCB and it does not copy the Korg drawing. Confirmed facts are cited by sheet id from 01-research.md. Numbers that the sheets do not print use a stand-in id from METHODOLOGY.md.

System

Stereo host buffers stop at the processor boundary. The graph runs in volts, one sample at a time, mono. Ext In makes the mono sum. Output turns volts back into the host's float range (S-01) and restores stereo with an explicit dry cable pair.

```
flowchart LR
    subgraph host [Host stereo]
        InL[In L]
        InR[In R]
        OutL[Out L]
        OutR[Out R]
    end
    subgraph rack [Fixed mono rack]
        ExtIn[Ext In]
        VCO[VCO]
        VCF[VCF]
        VCA1[VCA 1]
        VCA2[VCA 2]
        MG[MG]
        EG1[EG 1]
        EG2[EG 2]
        Noise[Noise]
        Div[Divider]
        Inv[Inverter]
        Lag[Integrator]
        RM[Ring]
        OutM[Output]
    end
    InL --> ExtIn
    InR --> ExtIn
    ExtIn --- VCO
    ExtIn --- VCF
    VCF --- VCA1
    VCA1 --- OutM
    OutM --> OutL
    OutM --> OutR
```

Lines inside the rack in that picture are not normalised connections. Only cables the user inserts, or the default patch inserts, are connections. The host boundary is not a cable.

Rack order, top to bottom, left to right:

1. Ext In, VCO, VCF, VCA 1, VCA 2, Output 2. MG, EG 1, EG 2, Noise, Ring 3. Divider, Inverter, Integrator

Phase 2 faces are not drawn and have no ports in the graph.

PatchGraph rules

- Fixed module list. No add-module command in phase 1.
- Each module owns a fixed `float portvalue[]`. Unpatched Audio and CV inputs are set to 0 V before the sample, except EG trig jacks. An unpatched trig is set to +5 V so it rests released. 0 V on a trigger means "shorted," and a missing cable must not hold the envelope forever (S-15). A Gate input that received a promoted Gate uses the promotion rule below.

- One cable per input. A second connect returns false and changes nothing.
- An output may feed several inputs (S-27). That fan-out is a software convenience so EG 1 can reach both the VCA and the filter before the multiples module exists.
- Cable creation and deletion happen on the message thread. The audio thread reads a published snapshot.
- Snapshot storage is two preallocated graphs plus an atomic index. Swapping snapshots does not allocate. processBlock does not allocate, lock, take a mutex, or log.
- Audio thread order is a topological order computed when the snapshot is published.
- Until step 19, a cable that would close a cycle is rejected. From step 19, a cycle is accepted: the newest cable in the cycle is the back-edge (S-24) and carries a one-sample delay. Every other cable in that snapshot is zero-delay.
- Port types: Audio, CV, Gate.
- Allowed: Audio to Audio, Audio to CV, CV to CV, CV to Audio, Gate to Gate, Gate to CV, Gate to Audio.
- Rejected: Audio or CV into a Gate-only input. Two cables into one input. A cable from a port to itself with no module in between.
- Gate promotion into a CV or Audio input uses S-15: held gate writes 0 V, released gate writes +5 V. That matches an active-low S-trig electrical picture. Logical "held" inside an EG still means the trigger condition is true.
- EG Trig jacks are type CV, not Gate-only, because the hardware detector accepts a voltage. Ext In Gate is type Gate and may be patched into Trig through the promotion above. MG pulse is type CV (0 V to +5 V) and may be patched into Trig directly. Divider outputs are type CV.
- Gate-only inputs in phase 1: none on the modules. Ext In's gate output is the Gate source. EG 2 DelayTrig is type Gate. Output has no gate jack.
- Sample rate lives on the graph. prepare(sampleRate) runs on the message thread before audio starts, and again if the host changes rate. It may allocate. processSample may not.

Sketch, not a finished class:

```
enum class PortType { Audio, CV, Gate };
enum class PortDir { In, Out };

struct PortDesc { const char* name; PortType type; PortDir dir; };

class Module {
public:
    virtual ~Module() = default;
    virtual int numPorts() const = 0;
    virtual PortDesc port(int index) const = 0;
    virtual void setKnob(int knob, float zeroToOne) = 0;
    virtual void prepare(double sampleRate) = 0;
    virtual void processSample() = 0; // reads and writes portValue only
    float portValue[8] {};
    double sampleRate = 48000.0;
};
```

processSample on a module reads portValue for inputs and writes portValue for outputs. The graph copies cable sources into destinations before each module runs. A delayed cable copies the previous sample.

Shared formulas

Volts are the graph unit. Knob widgets store 0 to 1 and the module maps them.

Exponential time or frequency knob:

$$\text{value} = \text{min} * \text{pow}(\text{max} / \text{min}, \text{knob}01)$$

One-pole smoother, used by VCA 2 and by the Ext In envelope:

$$\begin{aligned} \text{coeff} &= 1 - \exp(-1 / (\text{tauSeconds} * \text{sampleRate})) \\ \text{state} &+= (\text{target} - \text{state}) * \text{coeff} \end{aligned}$$

A held DC input to the integrator settles at the input. A module that ramps away on a held input is the wrong integrator.

Denormals: add nothing. If a one-pole state falls below $1e-15$ V, flush it to 0. Do this inside the module, with no heap.

NaN policy: Hz/V clamps at 0.05 V (S-03). Dividers do not divide by a signal. If a state becomes non-finite, replace that state with 0 at the end of processSample and count it on a preallocated uint32 for the test build only. The release audio thread does not print.

Default patch

Cables, in creation order:

#	From	To
1	Ext In Mono	VCF SigIn
2	VCF SigOut	VCA 1 SigIn
3	VCA 1 Out	Output Wet
4	Ext In L	Output L
5	Ext In R	Output R
6	Ext In Gate	EG 1 Trig
7	EG 1 OutA	VCA 1 Env
8	EG 1 OutA	VCF Cutoff

Cable 8 is fan-out (S-27). Mix defaults to 1 (fully wet). A silent input does not open VCA 1, so the wet output is silent until the gate stand-in fires. That is the intended reading of a VCA with no initial gain.

Ext In

Purpose: plugin boundary. Not an MS-50 module. Provides the stereo pair, a mono sum for the mono rack, and a gate so EG 1 can run before phase 2 TRIG SW and ESP exist.

Knobs:

Knob	Range	Default	Status
Gate threshold	0.05 V to 2 V on the follower	0.2 V	S-22
Gate release	10 ms to 500 ms	80 ms	S-22

The manual button is a momentary UI control, not a knob and not a module. It ORs a held gate while the mouse is down.

Jacks:

Jack	Dir	Type	Notes
L	out	Audio	Host left, in volts: sample * 5, so a full-scale host sample is ± 5 V (S-01 inverse at the input). STAND-IN input sensitivity
R	out	Audio	Host right, same scale
Mono	out	Audio	$(L + R) / 2$
Gate	out	Gate	1 while the button is down or the follower is above threshold

Signal flow: host float to volts, mono sum, absolute value into a one-pole follower (attack 5 ms stand-in, release from the knob), compare to threshold.

Failure modes: full-scale stereo summing to more than ± 5 V. Do not clip the mono sum in Ext In. Let the filter see it. DC from the host passes. That is acceptable.

Test: testExtInMonoAveragesStereo. Left +1 V-equivalent, right -1, mono 0. testExtInGateFiresAboveThreshold. testExtInButtonForcesGate.

CONFIRMED: nothing inside this block is an MS-50 circuit. STAND-IN: S-22, and the ± 5 V input scaling tied to S-01.

Output

Purpose: plugin boundary. Not the headphone amplifier.

Knobs:

Knob	Range	Default	Status
Mix	0 dry to 1 wet	1	S-23
Level	0 to 1, linear	0.8	S-23

Jacks: L in Audio, R in Audio, wet in Audio. No outputs inside the graph. The processor reads the module's outL and outR after the sample.

```
wet = Wet port (mono volts)
outL = (L * (1 - mix) + wet * mix) * level * 0.2
outR = (R * (1 - mix) + wet * mix) * level * 0.2
```

0.2 is S-01 (kOutputVoltsToUnit). Unpatched Wet is 0 V. Unpatched L/R are 0 V, so mix 0 with no dry cables is silence. The default patch inserts the dry cables.

Failure modes: Wet is mono, so a stereo image collapses as mix rises. That is specified. Do not build a stereo VCF to "fix" it.

Test: testDryMixPassesStereo at mix 0. testWetMixIgnoresDry at mix 1. testLevelZeroIsSilence.

VCO

Purpose: one oscillator, three outputs at once. Confirmed control topology from KOD-A40044. The transistor reset and the +18 V rail are not simulated in steps 0 to 20.

Knobs:

Knob	Range	Default	Status
Scale	32', 16', 8', 4'	8'	CONFIRMED positions. Hz in the table in METHODOLOGY are S-03
Freq A amount	0 to 1	0	S-05
Freq B amount	0 to 1	0	S-05
Fine	-1 to +1, widget centered	0	S-04, ± 2 semitones
PW	0 to 1	0.5	S-06

Jacks:

Jack	Dir	Type
Hz/V	in	CV
Oct/V	in	CV
FreqA	in	CV
FreqB	in	CV
PWM	in	CV
Saw	out	Audio
Tri	out	Audio

Jack	Dir	Type
Pulse	out	Audio

Signal flow, per sample:

```
footage = hzFor(scale)                # S-03 table
fine = pow(2, (fineKnob * 2) / 12)    # S-04, ±2 semitones
oct = Oct/V volts + FreqA*amountA + FreqB*amountB
octExpo = pow(2, oct)                 # S-04, S-05: 1 V = 1 octave at full amount
hzFromLinear = footage * max(Hz/V volts, 0.05) / 1.0
# Scale switch changes footage and therefore the linear law only.
hz = hzFromLinear * fine * octExpo
```

When Hz/V is unpatched it reads 0 V, and the clamp makes the linear term $\text{footage} * 0.05$, which is too low to be the unpatched panel pitch. Correction, still a stand-in: if the Hz/V jack has no cable, the linear term is footage (the reference pitch). If a cable is connected, S-03 applies, including the 0.05 V floor. The graph must tell the module whether the jack is patched. Add `bool inputConnected[8]` set by the graph each sample. No allocation.

Anti-alias: PolyBLEP saw, triangle from the integrated saw, pulse by comparing the saw phase to the duty. This anti-alias method is part of S-02, not a confirmed circuit.

Duty:

```
duty = clamp(0.05 + pwKnob * 0.90 + pwmVolts / 5 * 0.5, 0.05, 0.95)
```

Outputs: saw, triangle, and pulse each scaled to ± 5 V (S-02).

Failure modes: patching a 1 V/octave keyboard into Hz/V will not play octaves. That is correct. Patching Hz/V into Oct/V will not play Korg octaves. Also correct. Do not add sync.

Test: `testScaleDoesNotChangeOctJack` (same Oct/V voltage, two scale positions, the ratio of frequencies equals the footage ratio only through the linear term; with Hz/V unpatched, changing scale changes pitch by the footage ratio). `testOctIsOneVoltPerOctave`. `testHzPerVoltIsLinear`. `testThreeOutputsAlwaysRun`. `testPwmMovesDutyNotPitch`.

CONFIRMED: two pitch jacks, scale on the linear chain, three simultaneous waves, separate PWM, no sync. STAND-IN: S-02, S-03, S-04, S-05, S-06, PolyBLEP.

VCF

Purpose: the lowpass. Step 8 is a stable 2-pole (S-07, S-08, S-09). Step 20 replaces the cutoff computation and the feedback clip with the diode-bridge stand-in (S-10). Jacks do not change between those steps.

Knobs:

Knob	Range	Default	Status
Cutoff	0 to 1, exponential Hz in step 8	0.55	S-07
Peak	0 to 1	0.15	S-09 then S-10
Cutoff amount	0 to 1 attenuator on the CV jack	0.4	CONFIRMED that an attenuator exists. Depth S-08

Jacks: SigIn Audio in, Cutoff CV in, SigOut Audio out.

Step 8 flow:

```
cv = Cutoff volts * amount            # ±5 V typical
# Positive CV raises cutoff. Map ±5 V to ±4 octaves around the knob. S-08.
hz = knobHz * pow(2, cv / 5 * 4)
hz = clamp(hz, 15, 20000)
Q = 0.5 + peak * 7.5                  # S-09, Q ≤ 8, stable
```

Transposed direct form II biquad, lowpass, $w_0 = 2\pi \cdot \text{hz} / \text{sampleRate}$, $\alpha = \sin(w_0) / (2Q)$. Flush denormals. AC character: step 8 may pass DC. Step 20 adds a one-pole highpass at 5 Hz in front of the output as a stand-in for the output coupling capacitor. Mark that highpass S-10b in the code comment next to S-10.

Step 20 additions, still not a WDF and not a claim of measured CA3019 parameters:

- Compute a control current from cutoff and CV using the paper's shape $I_b = I_s * (\exp(v_{Bias} / (n * V_T)) - 1)$ with S-10's I_s and n . Map I_b to the two resistances of the linearized bridge the way the paper reduces the bridge to two resistors, using a precomputed function of bias, not a runtime matrix allocate.
- Run the 2-pole with those resistances.
- Apply tanh in the feedback path as the stand-in for D5 to D12.
- A larger $|SigIn|$ may reduce the effective bias slightly so cutoff and resonance fall. The coefficient of that pull is S-10 and starts small enough that a -12 dBFS saw does not mute the filter. Do not encode 250 Hz or 200 Hz.

Failure modes: Q above 8 in step 8. Hard-coded self-oscillation frequency. Using a Korg-35 or OTA formula. A highpass mode (that is the AM8319, not this sheet).

Test step 8: testVcfPassesDcOrLow (a 30 Hz sine is louder than a 15 kHz sine at low cutoff). testVcfPeakIncreasesResonance (ring longer at peak 1 than peak 0, and the output stays finite). testVcfPositiveCvRaisesCutoff.

Test step 20: testVcfStaysFiniteWhenDrivenHard. testVcfHotInputMovesSpectrum (a loud input and a quiet input do not produce the same normalized spectrum at high peak). testVcfHasNoHighpassSwitch.

CONFIRMED: 2-pole lowpass, CA3019 bridge as the target, peak knob, cutoff attenuator, clipper diodes exist, output coupling exists, not keyboard-tracked by a printed law. STAND-IN: S-07, S-08, S-09, S-10.

VCA 1

Purpose: audio VCA, sheet name VCA. Linear gain is S-11 until a later pass replaces it with a diode-bridge gain. Low-cut stays manual.

Knobs:

Knob	Range	Default	Status
Low cut	10 Hz to 2 kHz, exponential	10 Hz	S-11. The control exists: CONFIRMED
Intensity	0 to 1	0.85	CONFIRMED as an output attenuator. The taper is linear stand-in

Jacks: SigIn Audio in, Env CV in, out Audio out.

```
envGain = clamp(Env volts / 5, 0, 1)    # negative env closes. S-11
filtered = onePoleHighpass(SigIn, lowCutHz)
Out = filtered * envGain * intensity
```

No initial gain. Unpatched Env is 0 V, so the output is 0. AC coupling stand-in: the low-cut at its minimum (10 Hz) is the only highpass. Do not add a second DC block.

Failure modes: treating Intensity as a CV-depth knob. A CV jack on low-cut. Sharing this code path with VCA 2.

Test: testVca1SilentWithoutEnv. testVca1IntensityScalesOutput. testVca1LowCutDarkens (high low-cut reduces a 40 Hz sine and leaves a 2 kHz sine). testVca1NegativeEnvIsClosed.

CONFIRMED: manual low-cut, intensity after the gain cell, env jack, AC-coupled role. STAND-IN: S-11 linear gain.

VCA 2

Purpose: modulation VCA, sheet name MVCA. DC-coupled. Opto lag is a smoother (S-12).

Knobs: none.

Jacks: In CV in (also accepts Audio), Control CV in, out CV out.

```
target = clamp(Control volts / 5, 0, 1)
gain = onePole(target, tau = 0.020)    # S-12
```

Out = In * gain

A DC input of +3 V with control +5 V settles near +3 V. A pulse on Control does not pass instantly. 20 ms is the stand-in, not a measurement.

Failure modes: AC coupling. A panel knob that the sheet does not have. Reusing VCA 1's low-cut.

Test: testVca2PassesDc. testVca2ControlDoesNotClick (a step on Control produces a ramp longer than one sample). testVca2NoKnobs.

CONFIRMED: opto topology, DC path, no panel knob. STAND-IN: S-12.

MG

Purpose: modulation generator. UI title MG. Class name MgModule. Do not label the faceplate only "LFO".

Knobs:

Knob	Range	Default	Status
Frequency	0.01 Hz to 200 Hz, exponential knob	5 Hz	Endpoints CONFIRMED. Taper S-16
PW	0 to 1	0.5	One knob drives symmetry and pulse width. CONFIRMED wiring. Map S-16

Jacks: FreqMod CV in, PWM CV in, Tri Audio out, SawUp Audio out, SawDown Audio out, Pulse Audio out.

There is no FM amount knob on the MG sheet, only the jack. Stand-in depth, still S-16: a FreqMod voltage shifts frequency in hertz, not in octaves (that linear reading is INFERRED).

hz = clamp(knobHz + (FreqMod volts / 5) * knobHz, 0.01, 200)

Full-scale +5 V doubles the knob frequency. Full-scale -5 V reaches the 0.01 Hz clamp when the knob is the only other term. Call the scale factor out in a comment on the S-16 constants. Do not add an FM-amount knob.

sym = clamp(pwKnob + pwmVolts / 5, 0, 1)

sym 0: falling saw on the triangle output's shape, pulse duty 0.05. sym 0.5: triangle, duty 0.5. sym 1: rising saw, duty 0.95. The dedicated saw jacks always output the two slopes at ± 2.5 V, so a patch can pick a saw without turning PW. Pulse is 0 V to +5 V, never negative. Triangle and saws are ± 2.5 V. Those levels are CONFIRMED.

Failure modes: a bipolar pulse. A frequency knob that stops at 20 Hz. Separate knobs for symmetry and width.

Test: testMgPulseIsUnipolar. testMgTriangleIsBipolar2V5. testMgFreqEndpoints. testMgPwAffectsPulseAndTriangle. testMgSawJacksOpposite.

EG 1

Purpose: ADSR. Three outputs.

Knobs: Attack, Decay, Sustain, Release. Times S-13, 1 ms to 10 s exponential. Sustain 0 to 1. Defaults: A 0.01 s, D 0.25 s, S 0.6, R 0.30 s. Map the 0 to 1 widgets with S-13.

Jacks: Trig CV in, OutA CV out, OutB CV out, OutC CV out.

Trigger: held while Trig volts < 1.5 (S-15). Unpatched, the graph writes +5 V and the envelope stays idle. A promoted Gate writes 0 V when held, so Ext In Gate opens the EG, and writes +5 V when released. Rising edge (released to held) restarts attack. While held after decay, output sits at sustain. Release starts when the input goes above 1.5 V. Retrigger from a new edge during release restarts attack from the current level (no forced drop to 0). That edge rule is part of S-15.

Shapes, exponential segments toward the target (RC stand-in, S-13):

OutA: 0 V idle, up to +5 V, down to sustain*5 V, release toward 0 V
OutB: negation of OutA
OutC: OutA - sustain*5 V, so a sustain of 0.5 sits at 0 V and the peak is +5 V only when sustain is 0

OutC's centering is S-14 and is the first EG1 voltage to replace if a manual turns up. OutA and OutB's 5 V span is also S-14, chosen to match the EG2 cartoons.

Failure modes: one output. MS-20 EG1 timing (delay then attack then release) on this module. A sustain that is a time rather than a level.

Test: testEg1SustainLevel. testEg1OutBIsNegation. testEg1ReleasesWhenTriggerLifts. testEg1HasDecay. testEg1ThreeJacks. testEg1UnpatchedTrigIsIdle (no cable, Trig forced to +5 V, envelope stays at 0).

CONFIRMED: four ADSR knobs, three jacks, one trig. STAND-IN: S-13, S-14, S-15.

EG 2

Purpose: hold, delay, attack, release. No decay knob. No sustain knob. No sustain plateau in the stand-in.

Knobs: Hold, Delay, Attack, Release. All S-13, 1 ms to 10 s. Defaults: hold 1 ms, delay 1 ms, attack 0.02 s, release 0.20 s.

Jacks: Trig CV in (same 1.5 V rule, S-15), outPos CV out, outNeg CV out, DelayTrig Gate out.

S-26 timeline, which is the implementable spec:

1. On the edge into "held," set phase to Wait and clear the envelope to 0 V. 2. Wait for holdSeconds + delaySeconds. The envelope stays at 0 V. This is the stand-in. A later measurement may show that hold instead stretches a plateau. 3. At the end of the delay portion (time == holdSeconds), DelayTrig is high for one sample on the Gate output. The remaining wait is the rest of delay. If delay is at minimum, the pulse is at the end of hold. 4. Attack: rise from 0 V to +5 V in attack seconds. 5. Release starts as soon as attack finishes. Fall to 0 V. There is no flat top. 6. outNeg = -outPos. 7. A new edge during the cycle restarts at step 1.

DelayTrig is a Gate: high for that single sample, low otherwise. It can be patched into EG 1 Trig via Gate-to-CV promotion (writes 0 V for that sample, which is "held" under S-15). One sample may be too short for a detector that looks at edges if the edge requires two samples. Implementation rule: hold DelayTrig high for $\max(1, \text{round}(0.001 * \text{sampleRate}))$ samples (1 ms). That width is part of S-26.

Failure modes: adding Decay or Sustain knobs. Copying EG 1. A plateau at +5 V while the gate is held.

Test: testEg2HasNoSustainKnob. testEg2ReturnsToZeroWithoutAPlateau. testEg2DelayTrigAfterHold. testEg2NegIsNegation. testEg2Restart.

CONFIRMED: four knobs, no decay, no sustain, two opposite 5 V cartoons, delay trig jack exists. STAND-IN: S-26 timeline, S-13 times, 1 ms pulse width.

Noise

Purpose: white and pink. No panel knob. KOD-A40042.

Jacks: white Audio out, Pink Audio out.

White: xorshift32 inside the module, summed three-uniform approximation, scaled to about ± 2.5 V peak (S-17). Pink: Paul Kellett filter on that white noise, then the same peak scale. Seed is 0xA341316C at prepare, so tests are repeatable. Do not call rand().

The selected transistor and the internal trim are not controls. Do not add a level knob.

Failure modes: a panel level. Pink that is only a one-pole lowpass with no 3 dB/octave attempt. Allocating a noise buffer.

Test: testNoiseBothJacksMove. testNoiseSeedRepeats. testNoisePinkIsDarkerThanWhite (integrated absolute slope, or a Goertzel at 100 Hz versus 10 kHz: pink has a smaller high/low ratio than white). testNoiseHasNoKnobs.

CONFIRMED: two outputs, no panel knob, pink is filtered white. STAND-IN: S-17 spectrum and level.

Divider

Purpose: clock divider. $\div 2$ and $\div 4$ only. KOD-A40041.

Knobs: none.

Jacks: In CV in (Audio may be patched in), Div2 CV out, Div4 CV out.

Schmitt (S-18): go high when input crosses +0.5 V upward, go low when it crosses +0.3 V downward. On each rising output of the Schmitt, toggle div2. On each rising edge of div2, toggle div4. High level +5 V, low level 0 V.

An input that never crosses the window produces constant lows. A bipolar triangle of ± 2.5 V at audio rate does cross it. A tiny CV around 0.1 V does not. That matches the research note that a small CV will not clock the comparator.

Failure modes: a $\div 8$ output. Sine sub-oscillators. A frequency knob.

Test: testDividerOnlyTwoAndFour. testDividersSquareCounts (100 rising edges in, 50 rising edges on div2, 25 on div4). testDividerIgnoresTinySignal.

CONFIRMED: $\div 2$, $\div 4$, comparator front end. STAND-IN: S-18.

Inverter

Purpose: gain -1. KOD-A40043. DC-coupled.

Knobs: none. Do not expose the internal offset trim (S-19 forces offset 0).

Jacks: In CV in, out CV out.

Out = -In

Test: testInverterNegatesDc. testInverterNegatesAudio. testInverterHasNoKnobs.

CONFIRMED: unity inversion, DC path, no panel offset. STAND-IN: S-19 zero offset.

Integrator

Purpose: lag. Faceplate title Integrator. It stays a module. KOD-A40039.

Knobs:

Knob	Range	Default	Status
Time	1 ms to 2 s, exponential tau	50 ms	S-20

Jacks: In CV in, out CV out.

```
coeff = 1 - exp(-1 / (tau * sampleRate))
state += (In - state) * coeff
Out = state
```

Same sign. A constant +4 V input ends at +4 V. A fast Time dulls audio. A slow Time glides pitch when patched into Oct/V. It does not resonate and it does not run away.

Failure modes: hiding this inside the VCO as "glide." A free integrator. A resonant filter.

Test: testIntegratorSettlesToInput. testIntegratorSameSign. testIntegratorSlowIsSlower. testIntegratorIsItsOwnModule (the rack list contains it).

CONFIRMED: IN, TIME, OUT, lag behavior rather than a runaway ramp. STAND-IN: S-20 tau range and the one-pole shape.

Ring

Purpose: four-quadrant multiplier, RC4200 on KOD-A40040. Not the filter diodes.

Knobs: none. Bleed constant is 0 (S-21). Do not add a panel null.

Jacks: A CV in, B CV in, out CV out. Audio may be patched to A or B.

```
Out = (A * B) / 5 + bleed  
bleed = 0
```

+5 V and +5 V produce +5 V. +5 V and -5 V produce -5 V. Either input at 0 V produces 0 V. DC passes, so an envelope on B and audio on A is a VCA.

Failure modes: a diode-ring model shared with the VCF. AC coupling. A panel knob.

Test: `testRingFourQuadrant`. `testRingZeroKills`. `testRingPassesDcProduct`. `testRingHasNoKnobs`.

CONFIRMED: four-quadrant IC, two inputs, one output, no panel knob, DC path. STAND-IN: S-21 scale and zero bleed.

Phase 2 faces (documented, no ports)

Do not instantiate these in `ModuleRack`.

- Adding amp: three attenuators, inverting mix, DC. Needed later if fan-out (S-27) is removed.
- Sample and hold: IN, OUT, EXT CLOCK, CLOCK OUT, rate pot. FET hold. Not a stand-in inside Noise.
- ESP: preamp, follower, trigger. Not the Ext In gate (S-22), which must be deleted or relabeled when ESP exists.
- TRIG SW: active-low button module. Replaces the Ext In manual button's role as the obvious trigger.
- Volt source, meter, headphone amp: as in the research summary.
- Multiples: do not invent buffering. The sheet set does not show the circuit.

State

Step 18 stores: format version, sample-rate-independent knob values (0 to 1 or the scale index), and the cable list as module-id plus port-index pairs. It does not store module state (EG phase, filter memory, noise seed). Loading a preset calls `prepare` memory clear. Unknown version: reject the load, keep the current patch, report an error on the message thread.

What the audio thread is forbidden to do

Allocate, lock, read the message-thread cable vector, call into JUCE, resize a buffer, build a path for a cable, or walk the module list in UI order if that differs from the published topological order.

Build guide

Date: 2026-09-21.

Implement one step, run its acceptance test, then stop. A later step that seems small is still a later step. If the test fails, fix it inside the same step. If the fix needs a change to `METHODOLOGY.md` or `SCHEMATICS.md`, make that change in the same branch and say why. Do not silently retune a stand-in.

Branch name: `feat/step-xx` with `XX` from 00 to 20. Open it from `main` after the previous step is merged.

Global ban, every step: no MIDI note path, no neural net, no phase 2 module, no second copy of a phase 1 module, no heap use in `processBlock` or `processSample`, no Korg drawing scans.

JUCE is fetched at tag `8.0.4` unless that tag will not configure, in which case stop and record the tag that did. Do not vendor the JUCE tree.

Step 0: repo, docs, license

Files: `README.md`, `METHODOLOGY.md`, `LICENSE`, `.gitignore`, `docs/01-research.md`, `docs/SCHEMATICS.md`, `docs/BUILD_GUIDE.md`, `docs/TESTPLAN.md`, `docs/REPO_SETUP.md`, `docs/MS50_Modular_Design_Pack.pdf`, `scripts/init-repo.sh`, `scripts/build_design_pdf.py`.

Acceptance: `python3 scripts/build_design_pdf.py` exits 0 and the PDF opens. `git status` shows no audio files and no schematic scans. A search of the markdown finds no second stand-in number for a quantity that already has an id.

Do not touch: nothing exists yet besides these files. Do not add `CMakeLists.txt`.

Rollback: delete the commit on the design branch before it is pushed. After it is pushed, revert the commit. Do not rewrite `main` if anyone else has it.

Step 1: empty JUCE FX plugin builds

Files to create:

- `CMakeLists.txt` as below
- `Source/PluginProcessor.h`
- `Source/PluginProcessor.cpp`
- `Source/PluginEditor.h`
- `Source/PluginEditor.cpp`

`CMakeLists.txt`:

```
cmake_minimum_required(VERSION 3.22)
project(MS50Modular VERSION 0.1.0 LANGUAGES C CXX)
set(CMAKE_CXX_STANDARD 20)
set(CMAKE_CXX_STANDARD_REQUIRED ON)

include(FetchContent)
FetchContent_Declare(
  JUCE
  GIT_REPOSITORY https://github.com/juce-framework/JUCE.git
  GIT_TAG 8.0.4
  GIT_SHALLOW TRUE)
FetchContent_MakeAvailable(JUCE)

juce_add_plugin(MS50Modular
  COMPANY_NAME "Personal"
  IS_SYNTH FALSE
  NEEDS_MIDI_INPUT FALSE
  NEEDS_MIDI_OUTPUT FALSE)
```

```

IS_MIDI_EFFECT FALSE
EDITOR_WANTS_KEYBOARD_FOCUS FALSE
COPY_PLUGIN_AFTER_BUILD FALSE
PLUGIN_MANUFACTURER_CODE Psnl
PLUGIN_CODE Ms50
FORMATS VST3
PRODUCT_NAME "MS-50 Modular")

target_sources(MS50Modular PRIVATE
    Source/PluginProcessor.cpp
    Source/PluginEditor.cpp)

target_compile_definitions(MS50Modular PUBLIC
    JUCE_WEB_BROWSER=0
    JUCE_USE_CURL=0
    JUCE_VST3_CAN_REPLACE_VST2=0)

target_link_libraries(MS50Modular PRIVATE
    juce::juce_audio_utils
    juce::juce_recommended_config_flags
    juce::juce_recommended_warning_flags)

```

Processor: stereo in, stereo out, isBusesLayoutSupported accepts only stereo. processBlock copies input to output.
 Editor: a label "MS-50 Modular" and the words "no patch yet".

Acceptance:

```

cmake -S . -B build -DCMAKE_BUILD_TYPE=Debug
cmake --build build --config Debug

```

The VST3 bundle exists under build/. Load it in a host. Stereo audio passes, bit-identical to the input aside from host float handling. MIDI notes do nothing because MIDI input is off.

Do not touch: docs, except a one-line "step 1 builds" note if the JUCE tag had to change.

Rollback: remove CMakeLists.txt and Source/ and the build/ directory. build/ is gitignored.

Step 2: Port, Cable, Module, PatchGraph

Files: Source/Modular/Port.h, Module.h, Module.cpp, Cable.h, PatchGraph.h, PatchGraph.cpp, Source/Tests/GraphTests.cpp, and a CMake target MS50ModularTests that is a console app, not a plugin. It links the modular sources and does not link the plugin target.

Implement the rules in SCHEMATICS.md up to, but not including, delayed feedback. A cycle is rejected. One cable per input. Fan-out allowed. Type matrix enforced. Two snapshots, atomic index, no alloc in a method named process.

Module is the base from the schematic sketch. No concrete synth modules yet. Tests use a GainModule in the test file only, gain fixed at 1, not on the rack.

Acceptance: MS50ModularTests runs testOneCablePerInput, testFanOutAllowed, testRejectSignalIntoGate, testRejectCycle, testSnapshotSwapDoesNotAllocate (the process function's source does not call new or push_back; a debug counter of allocations stays 0 across 1000 samples).

Do not touch: PluginProcessor audio behavior, the editor layout.

Rollback: delete Source/Modular and Source/Tests and the test target. The plugin from step 1 still passes audio.

Step 3: Ext In, Output, cable copy

Files: Source/Modular/ExtIn.h/.cpp, OutputModule.h/.cpp. Processor owns one PatchGraph, one Ext In, one Output. It converts host floats using S-01 and runs the graph.

Default cables for this step only: Ext In L to Output L, Ext In R to Output R. No wet cable yet. Mix 0.

Acceptance: `testDryMixPassesStereo` and `testExtInMonoAveragesStereo`. In a host, a stereo file played through the plugin at mix 0 matches the dry file within -90 dBFS residual. Mono material on the left only stays on the left.

Do not touch: cable drawing, VCF, any other module.

Rollback: processor returns to the step 1 copy. Keep the graph types.

Step 4: GUI rack and jacks, no DSP change

Files: `Source/UI/RackView.h/.cpp`, `JackView.h/.cpp`. Editor shows the 14 faceplates, all of them, including Integrator, even though only Ext In and Output process audio. Other faceplates are inert and their jacks do not connect yet. Each jack is a circle with a type color: audio amber, CV blue, gate white.

Acceptance: the plugin still passes the step 3 dry test. The window lists all 14 names from the README table. Resizing the editor does not move audio. A search shows `RackView` does not call `processSample`.

Do not touch: graph rules, Ext In math.

Rollback: editor back to the step 1 label. Keep the view files only if they compile out via the editor.

Step 5: animated cables

Files: `Source/UI/CableView.h/.cpp`. Draw one cubic from jack center to jack center for each cable in the published snapshot. Animation is a dash phase advanced by a `Timer` on the message thread, 30 Hz is enough. No timer on the audio thread. No allocation per frame: a `juce::Path` member is rebuilt only when endpoints move.

Acceptance: with the two dry cables, both cables are visible and the dashes move. CPU of the audio thread does not include `CableView` (it is not called from `processBlock`). Dry audio test still passes.

Do not touch: DSP modules.

Rollback: draw nothing. Cables still exist in the graph.

Step 6: click-drag patching

Files: `RackView` mouse handlers, `PatchGraph::connect / disconnect` called only from the message thread.

Drag from an output to an input. On illegal pairs, refuse and show a one-line status string: "that jack does not take this cable" or "input already has a cable". Right-click a cable to remove it. Creating a cycle shows "feedback is not available until step 19" and does not connect. That string changes in step 19.

Acceptance: `testOneCablePerInput` still passes when the UI path is used (call `connect` the way the UI calls it). A manual check: drag Ext In Mono onto Output Wet, then set mix to 1, and a mono sum is heard on both speakers. Drag a second cable onto Output Wet and the first remains.

Do not touch: module DSP that does not exist yet.

Rollback: ignore mouse-up. Keyboard of cables is not required.

Step 7: Noise

Files: `Source/Modular/Noise.h/.cpp`. Add it to the rack's process list. Faceplate already exists; wire the two jacks.

Acceptance: `testNoiseBothJacksMove`, `testNoiseSeedRepeats`, `testNoisePinkIsDarkerThanWhite`, `testNoiseHasNoKnobs`. Patch White to Output Wet, mix 1, and confirm the host meters move. No level knob on the faceplate.

Do not touch: VCF, VCO, graph rules.

Rollback: remove Noise from the process list. Leave the faceplate.

Step 8: VCF stand-in

Files: `Source/Modular/Vcf.h/.cpp`. Implement step 8 formulas only (S-07, S-08, S-09). Do not include S-10 diode code in this step. A comment at the top of `Vcf.cpp` says "STAND-IN step 8, replaced in step 20".

Wire knobs on the faceplate: Cutoff, Peak, Cutoff amount.

Cables that may exist at the end of this step: Ext In L to Output L, Ext In R to Output R, Ext In Mono to VCF SigIn, VCF SigOut to Output Wet. Cables that mention VCA 1 or EG 1 wait for steps 9 and 10. Do not claim the eight-cable default patch yet.

Acceptance: the three VCF tests in `SCHEMATICS.md`. A host tone through Mono, the VCF, and Output Wet is quieter when cutoff is 0 than when cutoff is 1, and it does not explode at peak 1. Residual after 2 seconds of a loud sine is finite. Mix 1 for that listen. Mix 0 still passes dry stereo on the L/R cables.

Do not touch: VCA law, MG, the graph cycle rule.

Rollback: bypass VCF (`SigOut = SigIn`) and say so in the editor status line. Do not leave a half-updated biquad.

Step 9: VCA 1 and VCA 2

Files: `Source/Modular/Vca1.h/.cpp`, `Vca2.h/.cpp`. Different classes. No shared "Vca" with a mode flag.

Rewire the wet path: VCF SigOut to VCA 1 SigIn, VCA 1 Out to Output Wet. Remove the direct VCF-to-Output cable from step 8.

Acceptance: all VCA tests in the schematic. The test harness sets VCA 1 Env to +5 V when it needs a tone. In the plugin, until step 10, Wet is silent because Env is unpatched at 0 V. That silence is correct. Do not add a secret initial gain to make it louder. Dry mix 0 still passes.

Do not touch: VCF formulas, Ext In gate math beyond what step 3 already did.

Rollback: VCA 1 output 0. VCA 2 output 0. Do not route around them with a hidden gain.

Step 10: EG 1

Files: `Source/Modular/Eg1.h/.cpp`. Wire the default cables that use OutA. Ext In Gate promotes into Trig.

Add the remaining default cables: Ext In Gate to EG 1 Trig, EG 1 OutA to VCA 1 Env, EG 1 OutA to VCF Cutoff. The patch now matches the eight-cable list in `SCHEMATICS.md`.

Acceptance: EG1 tests in the schematic. Manual: hold the Ext In button, a tone at the input, mix 1, and the sound follows attack and release. Release the button and the sound dies. OutB patched to a meter or to VCA 2 control in a throwaway test cable moves the opposite way. Remove that throwaway cable before committing. Unpatched Trig rests at +5 V and does not hold the envelope open.

Do not touch: EG 2, VCF diode work.

Rollback: leave VCA 1 Env unpatched (0 V) so the default patch goes silent rather than stuck open.

Step 11: MG

Files: `Source/Modular/Mg.h/.cpp`. Faceplate title MG.

Acceptance: MG tests in the schematic, including unipolar pulse and 0.01 Hz to 200 Hz. Listen: patch Tri to VCF Cutoff with the default OutA cable removed for the listen, then restore the default cables before commit. The listen is not the acceptance test. The test is.

Do not touch: VCO, EG 2.

Rollback: MG outputs 0. Frequency knob stays on the panel.

Step 12: VCO

Files: Source/Modular/Vco.h/.cpp. PolyBLEP stand-in. inputConnected flags from the graph, as the schematic requires, so an unpatched Hz/V jack does not fall to the 0.05 floor.

Acceptance: VCO tests in the schematic. Listen: saw into Output Wet, scale switch changes pitch, Oct/V from the volt-free test (a test-only +1 V) doubles frequency. Do not add a keyboard to perform that test.

Do not touch: filter code, MG waveform code (do not "share an oscillator class" if that merges their levels; MG is ± 2.5 V and the VCO stand-in is ± 5 V).

Rollback: VCO outputs 0. Keep the jacks.

Step 13: EG 2

Files: Source/Modular/Eg2.h/.cpp. No decay parameter. No sustain parameter. If a knob widget is added for either, the step fails.

Acceptance: EG2 tests, including no plateau and delay-trig timing. Patch DelayTrig to EG 1 Trig in place of Ext In Gate for a manual check, then restore the default patch.

Do not touch: EG 1 state machine. Do not generalize both EGs into one class with a mode enum. Two classes may share a private one-pole segment helper if it contains no knob policy.

Rollback: EG 2 outputs 0 and DelayTrig low.

Step 14: Ring

Files: Source/Modular/Ring.h/.cpp.

Acceptance: ring tests. Manual: Noise White into A, MG Tri into B, Out into Output Wet. The result is not the sum of the two inputs (sum would still show the MG alone if one input is silent; product goes quiet if either side is 0). Restore the default patch after listening.

Do not touch: VCF. Do not reuse ring code as the filter nonlinearity.

Rollback: ring output 0.

Step 15: Divider

Files: Source/Modular/Divider.h/.cpp.

Acceptance: divider tests. No $\div 8$ jack on the faceplate. Manual: VCO pulse into divider, Div2 into Output Wet, pitch drops one octave relative to the pulse. Pulse stand-in is bipolar ± 5 V (S-02), and the Schmitt still triggers because ± 5 V crosses ± 0.5 V. If it does not, fix the Schmitt window in S-18's constant, in this step, and record the edit.

Do not touch: VCO pitch laws.

Rollback: both divider outputs 0.

Step 16: Inverter

Files: Source/Modular/Inverter.h/.cpp.

Acceptance: inverter tests. Manual: EG 1 OutA into the inverter into VCF Cutoff, with cable 8 removed. The filter moves the opposite way from OutA into the cutoff. Restore cable 8.

Do not touch: adding-amp code. Do not add an offset knob.

Rollback: output 0, which is wrong musically but safe. Prefer rollback to `out = In` only if a test name says so. This step's rollback is `out = 0` so a bad negate cannot ship as "wire".

Step 17: Integrator

Files: `Source/Modular/Integrator.h/.cpp`. The rack already shows the faceplate from step 4. This step makes it compute.

Acceptance: integrator tests, including settle-to-input and "module is on the rack." Manual: a fast MG saw through a slow integrator into Output Wet is duller than the dry saw. A + constant (use VCA 2 with a held gate-promoted control and a DC input from Ext In of a quiet DC, or a unit test) settles. The unit test is the acceptance. The listen is extra.

Do not touch: VCO fine tune. Do not move glide inside the VCO.

Rollback: `out = In` would hide a broken lag. Rollback is `out = 0` plus a status line "integrator bypassed to zero".

Step 18: state save and load

Files: `PatchGraph::getState / setState, processor getStateInformation / setStateInformation`.

Store version integer 1, every knob, the scale index, and cables as integer ids. Do not store filter memory or the noise seed. On load, clear module state via the existing prepare reset path.

Acceptance: `testPresetRoundTrip` builds the default patch, moves cutoff and adds one extra legal cable, saves, loads into a fresh graph, and sees the same knobs and cables. `testPresetRejectsBadVersion` keeps the previous patch. Host check: save a DAW project, reopen, cables match. The DAW project file is not committed.

Do not touch: DSP formulas.

Rollback: `getStateInformation` writes an empty block and `setState` no-ops. Document that presets will not survive.

Step 19: topological order and feedback delay

Files: `PatchGraph` sort and delay. Update the UI string from step 6.

Acceptance: `testFeedbackIsOneSample` builds a one-module cycle through a test gain of 1 (input + output cable loop). The delayed edge is the newest cable. A impulse returns on the next sample, not the same sample. `testNoAllocInProgress` still passes. Two legal feedback cables in one cycle: only the newest is delayed. An older cycle-free patch (the default eight cables) has zero delayed cables.

Do not touch: module formulas. Do not delay every cable "to be safe."

Rollback: reject cycles again, as in step 2. The UI string returns to the step 6 wording.

Step 20: filter closer to the diode bridge

Files: `vcf.cpp` only, plus the tests named for step 20. Update the file header to "step 20, S-10". Keep the jacks and knob ids stable so presets from step 18 still point at Cutoff, Peak, and Amount.

Implement the S-10 notes in the schematic: paper Is and n marked PAPER-SUBSTITUTE, tanh stand-in for D5 to D12, a small input-level pull, 5 Hz output highpass stand-in, no hard-coded 250 Hz.

Acceptance: step 8 tests that are still meaningful (`testVcfPositiveCvRaisesCutoff`, finite output) plus `testVcfStaysFiniteWhenDrivenHard` and `testVcfHotInputMovesSpectrum`. Listen to Gladén's video using the procedure in `METHODOLOGY.md`. Write `docs/listening/step-20.md` with the three lines the methodology asks for (matched, not matched, which id). That listening note is committed. Recordings are not.

Do not touch: VCO, VCA classes, EG classes, graph rules. Do not replace the VCF with a copied WDF framework.

Rollback: restore the step 8 biquad in `vcf.cpp` and say "S-10 reverted" in the listening note. Keep the step 20 tests compiling against the old behavior only if you mark them skipped. Do not delete the listening note.

After step 20

Stop. Phase 2 is a new packet. Do not start the adding amplifier because fan-out made it look easy.

Test plan

Date: 2026-09-21.

Automated tests live in `Source/Tests/GraphTests.cpp` and are run with the `MS50ModularTests` target from step 2 on. Host checks are done in a real plugin host after the step that needs them. This environment's design pack does not include a host, so a step that says "host" is not done until a person or an agent with a host has done it and written the host name.

Stand-in ids are defined in `METHODOLOGY.md`. A test that encodes a stand-in number must name the id in a comment on the assertion.

Plugin loads

After step 1:

- VST3 builds from the CMake command in the build guide.
- A host scans it as an effect, not an instrument.
- Stereo track, stereo out. A mono track is not a supported layout (`isBusesLayoutSupported` returns false).
- Playing MIDI notes does not change audio. MIDI input is off.
- Bypass in the host returns dry stereo, subject to the host's own bypass. This is a host feature, not an Output-module test.

Dry Ext In to Output

After step 3, and again after steps 6, 8, and 18:

- Cables: Ext In L to Output L, Ext In R to Output R. Mix 0.
- Left-only signal stays left. Right-only stays right.
- A file recorded through the plugin at mix 0 matches the source within a residual below -90 dBFS, delay 0 samples.
- Mix 1 with Wet unpatched is silence.
- `testDryMixPassesStereo` and `testExtInMonoAveragesStereo` pass.

Each module

Run the named tests when the matching build step lands. "Finite" means every sample in a 1 second render at 48 kHz is finite and inside ± 40 V in the graph (well above the ± 5 V stand-in, tight enough to catch a blow-up).

Module	Tests	Extra check
Ext In	<code>testExtInMonoAveragesStereo</code> , <code>testExtInGateFiresAboveThreshold</code> , <code>testExtInButtonForcesGate</code>	Button is on the Ext In faceplate, not a hidden key
Output	<code>testDryMixPassesStereo</code> , <code>testWetMixIgnoresDry</code> , <code>testLevelZeroIsSilence</code>	Stereo image collapses only as mix rises
Graph	<code>testOneCablePerInput</code> , <code>testFanOutAllowed</code> , <code>testRejectSignalIntoGate</code> , <code>testRejectCycle</code> until step 19, then <code>testFeedbackIsOneSample</code>	Second cable does not replace the first
Noise	<code>testNoiseBothJacksMove</code> , <code>testNoiseSeedRepeats</code> , <code>testNoisePinkIsDarkerThanWhite</code> , <code>testNoiseHasNoKnobs</code>	No level control on the panel

Module	Tests	Extra check
VCF step 8	testVcfPassesDcOrLow, testVcfPeakIncreasesResonance, testVcfPositiveCvRaisesCutoff	Peak at 1 stays finite. No highpass switch
VCF step 20	testVcfStaysFiniteWhenDrivenHard, testVcfHotInputMovesSpectrum	Listening note docs/listening/step-20.md exists. No constant 250
VCA 1	testVca1SilentWithoutEnv, testVca1IntensityScalesOutput, testVca1LowCutDarkens, testVca1NegativeEnvIsClosed	Unpatched env is silence, not unity
VCA 2	testVca2PassesDc, testVca2ControlDoesNotClick, testVca2NoKnobs	Different class from VCA 1
EG 1	testEg1SustainLevel, testEg1OutBIsNegation, testEg1ReleasesWhenTriggerLifts, testEg1HasDecay, testEg1ThreeJacks, testEg1UnpatchedTrigIsIdle	Three jacks visible. A missing trig cable does not hold the envelope
MG	testMgPulseIsUnipolar, testMgTriangleIsBipolar2V5, testMgFreqEndpoints, testMgPwAffectsPulseAndTriangle, testMgSawJacksOpposite	Faceplate says MG. Pulse never goes below 0 V
VCO	testScaleDoesNotChangeOctJack, testOctIsOneVoltPerOctave, testHzPerVoltIsLinear, testThreeOutputsAlwaysRun, testPwmMovesDutyNotPitch	Unpatched Hz/V still sounds at the footage pitch
EG 2	testEg2HasNoSustainKnob, testEg2 ReturnsToZeroWithoutAPlateau, testEg2DelayTrigAfterHold, testEg2NegIsNegation, testEg2Restart	Faceplate has no decay and no sustain
Ring	testRingFourQuadrant, testRingZeroKills, testRingPassesDcProduct, testRingHasNoKnobs	Product, not a sum
Divider	testDividerOnlyTwoAndFour, testDividerSquareCounts, testDividerIgnoresTinySignal	100 input edges, 50 and 25 output edges
Inverter	testInverterNegatesDc, testInverterNegatesAudio, testInverterHasNoKnobs	+3 V becomes -3 V
Integrator	testIntegratorSettlesToInput, testIntegratorSameSign, testIntegratorSlowIsSlower, testIntegratorIsItsOwnModule	Still a faceplate after every UI step

Default patch smoke, after step 10 and again after step 20:

- Eight cables from SCHEMATICS.md are present.
- Silence in, mix 1: silence out (VCA 1 closed).
- A 220 Hz sine at -12 dBFS, Ext In button held: signal appears after the attack and dies after release.
- Integrator faceplate is visible while this runs.

Illegal patches

Attempt	Result
Second cable into Output Wet	Rejected. First cable stays. Status string on the message thread
Audio or CV into EG 2 DelayTrig, which is Gate	Rejected
Gate into VCA 1 Env, which is CV	Allowed. Held gate writes 0 V (S-15), so the VCA closes while the gate is held. This is surprising and must be stated in the status line the first time it happens: "gate patched as active-low voltage"
Cable that closes a cycle, before step 19	Rejected
Cable that closes a cycle, after step 19	Accepted. Newest cable delayed one sample
Cable from a jack to the same jack	Rejected
Disconnect of a missing cable	No-op, no crash

UI and graph must agree. A test calls the same connect the mouse-up handler calls.

Preset recall

After step 18:

- testPresetRoundTrip and testPresetRejectsBadVersion.
- Host: save a project with a non-default cutoff and one extra cable. Reopen. Knobs and cables match. EG phase does not need to match. Noise seed does not need to match.
- A corrupt blob does not crash setStateInformation. The previous in-memory patch remains.
- Version field is an integer at the start of the block.

CPU smoke

After step 20, in a host, 48 kHz, buffer 64:

- Default patch, silence in, button not held, for 30 seconds. The audio thread stays free of allocations (plugin built with a debug allocator counter, or Instruments/Time Profiler showing no operator new on the audio thread).
- Default patch, -12 dBFS stereo noise in, button held, peak at 1, for 30 seconds. No dropouts on a machine that can run an empty VST3 at that buffer. This is a smoke test, not a cycle budget.
- Drag cables for 10 seconds while audio runs. No crash, no audio-thread lock. The snapshot index only changes on mouse-up.
- Feedback patch after step 19: VCF SigOut to a test gain of 0.5 and back into SigIn is not possible with one SigIn. Use Ring as a cycle: A and B fed from Out through one delayed cable and one fan-out, or the unit-test cycle. The host check is: a user-made cycle does not halt the audio thread for one buffer of 64.

If the smoke test glitches, the rollback is the build-guide rollback for the step that introduced the glitch, not a new thread inside processBlock.

Listening checks against public video

Do not commit the videos or the bounces.

When	Source	Pass condition
After step 8 and again after step 20	Gladén MwJk36KnB04, MS-50 sections at 0:51, 2:28, 4:20	Low peak is duller at low cutoff. High peak rings and stays finite. Step 20: a louder saw is not a scaled copy of a quiet saw
After step 12	Synth Party C6oqCY0ArvM after 0:51, as a reminder of Hz/V plus S-trig	Plugin Oct/V and Hz/V tests already encode the laws. The video is a sanity listen, not a pitch measurement
After step 11	Alex Ball 3YFDDDXw6F8 near 8:43	Optional. Confirms you are not listening to a Korg-35 story. Not a measurement

When	Source	Pass condition
Never as a timbre reference	Dr. Kunz _iJJYswUYd0, Perfect Circuit UXW39L0-bxY	Effects or a second synth are in the file

Write mismatches into docs/listening/step-20.md. Change a stand-in only in a follow-up commit that names the id.